

Integration by Parts



The integration by parts formula

- 1 Evaluate $\int x e^x dx$ using integration by parts.
 - 2 Evaluate $\int x \cos x dx$ using integration by parts.
 - 3 Evaluate $\int \ln x dx$ using integration by parts (with $dv = dx$).
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Repeated and definite applications

- 4 Evaluate $\int x^2 e^x dx$, applying integration by parts twice.
 - 5 Evaluate $\int_0^1 x e^{2x} dx$ exactly.
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Choosing u and dv wisely (LIATE)

- 6 Evaluate $\int x \ln x dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).
 - 7 Evaluate $\int x^2 \cos x dx$, applying integration by parts twice.
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The recurring-integral trick

- 8 Evaluate $\int e^x \sin x dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.
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Definite integrals by parts

- 9 Evaluate $\int_0^\pi x \sin x dx$ exactly.
 - 10 Evaluate $\int_1^e x^2 \ln x dx$ exactly.
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Reduction formulas

- 11** Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x dx = x^n e^x - n \int x^{n-1} e^x dx$.
- 12** Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x dx$, confirming it matches an earlier result.